

SECURITY FOR AI

Q2 2026

Unit PAM
Divisi Digital Product
Telkom Indonesia



WHY WE MUST SECURE AI

From Innovation Enabler to Critical Risk Vector

The Risk Reality



TOP GLOBAL RISK

Ranked as a **Top 6** long-term global risk by the World Economic Forum (2024).



CYBERCRIME-AS-A-SERVICE (CaaS)

Attack barrier lowered: Phishing and Deepfake kits now cost **<\$50/month**.

Business Impact

Case Study



THE REGULATORY HAMMER

Non-compliance with the EU AI Act can cost up to **7% of global turnover** (€35M+).



RESILIENCE = TRUST

“Digital resilience is the capacity to maintain public trust when systems fail or hallucinate.”

Standards & Frameworks

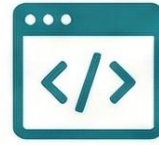
Global Benchmarks for AI Governance



ISO/IEC 42001

AI Management System

The global standard for managing organizational AI risks.



Technical Security

Engineering Baselines

- **OWASP Top 10 for LLMs:** The industry standard checklist for vulnerabilities (Injection, Theft, Supply Chain).
- **ISO/IEC DIS 27090:** Addresses specific threats like model poisoning.



NIST AI RMF

Risk Management Framework

A voluntary US framework to map, measure, and manage lifecycle risks.



Regulatory Benchmarks

Compliance Mandates

- **EU AI Act:** Risk-based classification system.
- **US Executive Order:** Mandates safety testing for dual-use foundation models.

AI Security Goals

Building Safe, Controlled, and Auditable AI



Data Protection

Prevent data and IP leaks from public GenAI use. No accidental sharing of sensitive company data.



Controlled Model Behavior

Detect and block prompt injection, jailbreaks, and toxic output. Real-time AI guardrails.



Secure AI Supply Chain

Only trusted models, weights, and libraries allowed. Unsafe formats and hidden code are blocked.



Continuous Assurance

Ongoing red teaming and monitoring. Security is continuous, not one-time.



From Policy to Practice

Real technical implementation, not just written policy. Ready to be shown as a working example.

AI Security Implementation

From Secure Access to Continuous Assurance

PHASE 1 – Infrastructure Hardening

PHASE 2 – Secure Model &

PHASE 3 – Product Guardrails

PHASE 4 – RAG & Data

PHASE 5 – Continuous Red



Control Plane & Secure Access

Output Name: Telkom AI Gateway with Policy Enforcement

What We Build:

- One secure entry point for all AI models and APIs
- Central policy control for access, cost, and usage
- Full audit trail for all AI requests

Main Deliverables

- AI Gateway reference implementation
- Authentication and authorization policy
- Rate limit and budget control rules
- Model allowlist and routing logic
- Access and usage logging standard

Adopt (Benchmark Tools):

- LiteLLM Proxy (gateway, usage tracking, budgets)
- Envoy Proxy with ext_authz
- API gateway alternatives (Kong, Apigee)

Build (Telkom-Specific):

- Risk tiers (Public / Controlled / Sovereign)
- Integration with Apigee as AI marketplace
- Local data residency enforcement



Guardrails & RAG Governance

Output Name: Telkom Guardrails Reference & RAG Security Playbook

What We Build:

- Controlled AI behavior at runtime
- Safe data access for RAG systems
- Protection from prompt injection and misuse

Main Deliverables

- Runtime guardrails design (input, tool, output)
- Guardrail policy templates (reject vs reword)
- RAG access control rules (RBAC propagation)
- Data sanitization and PII masking guidance
- Anti-indirect prompt injection controls

Adopt (Benchmark Tools):

- NeMo Guardrails (programmable guardrails)
- OWASP LLM Top 10 & agent safety guidance
- Vector DB security features (Elastic, OpenSearch, Pinecone)

Build (Telkom-Specific):

- Local language and cultural safety rules
- RAG enforcement integrated with enterprise IAM
- Governance rules aligned with Komdigi



Continuous Testing & Assurance

Output Name: AI Red Teaming Pipeline, Dashboard, and IR SOP

What We Build:

- Continuous validation of AI security risks
- Early detection of misuse and abuse
- Clear response when incidents happen

Main Deliverables

- Automated red teaming pipeline
- Initial test suite mapped to OWASP LLM Top 10
- AI security monitoring dashboard design
- AI misuse incident response SOP
- Feedback loop to update guardrails

Adopt (Benchmark Tools):

- Garak (LLM vulnerability scanner)
- PyRIT (automated GenAI red teaming)
- MITRE ATLAS (AI threat taxonomy)

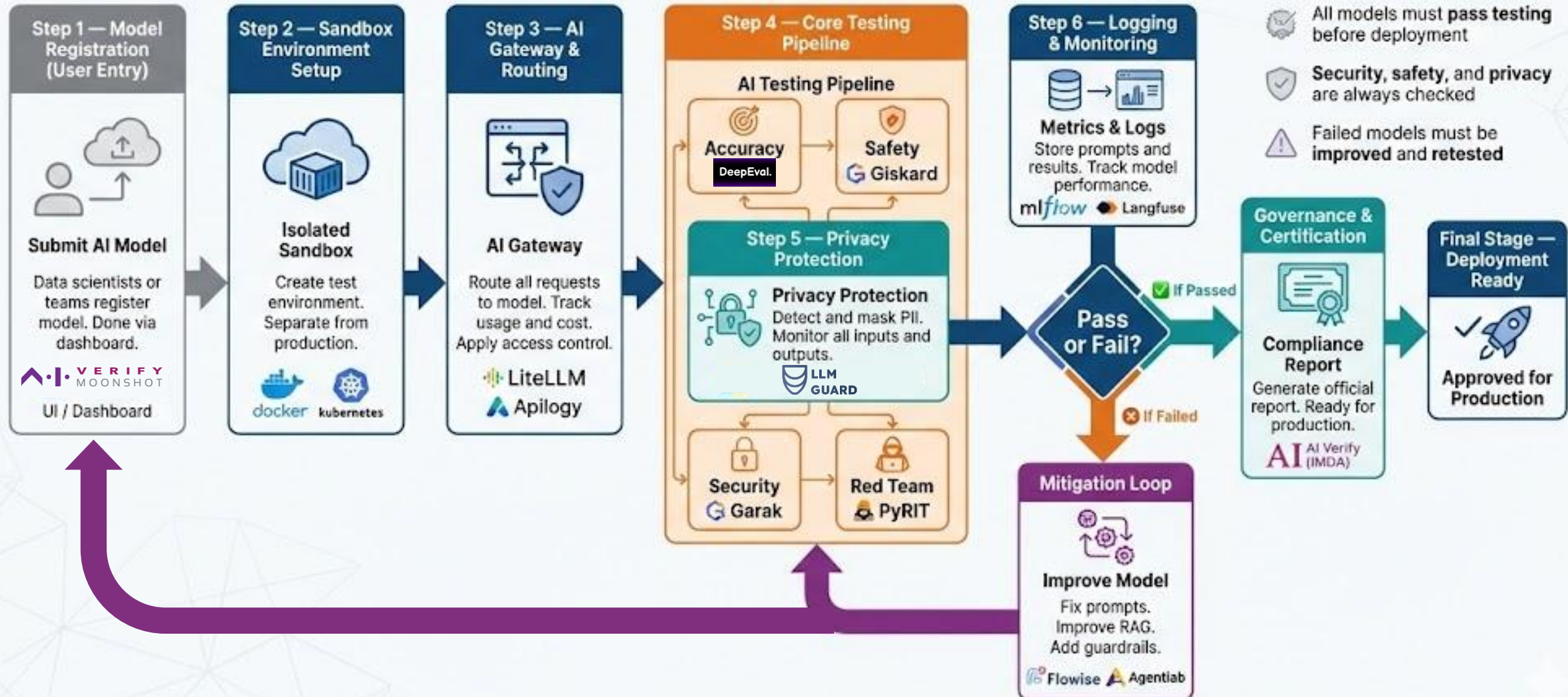
Build (Telkom-Specific):

- Custom attack scenarios for local context
- Integration with SOC and SIEM
- Local audit and reporting format

Together, these outputs deliver a secure, controlled, and auditable AI platform — from access control, to safe behavior, to continuous assurance.

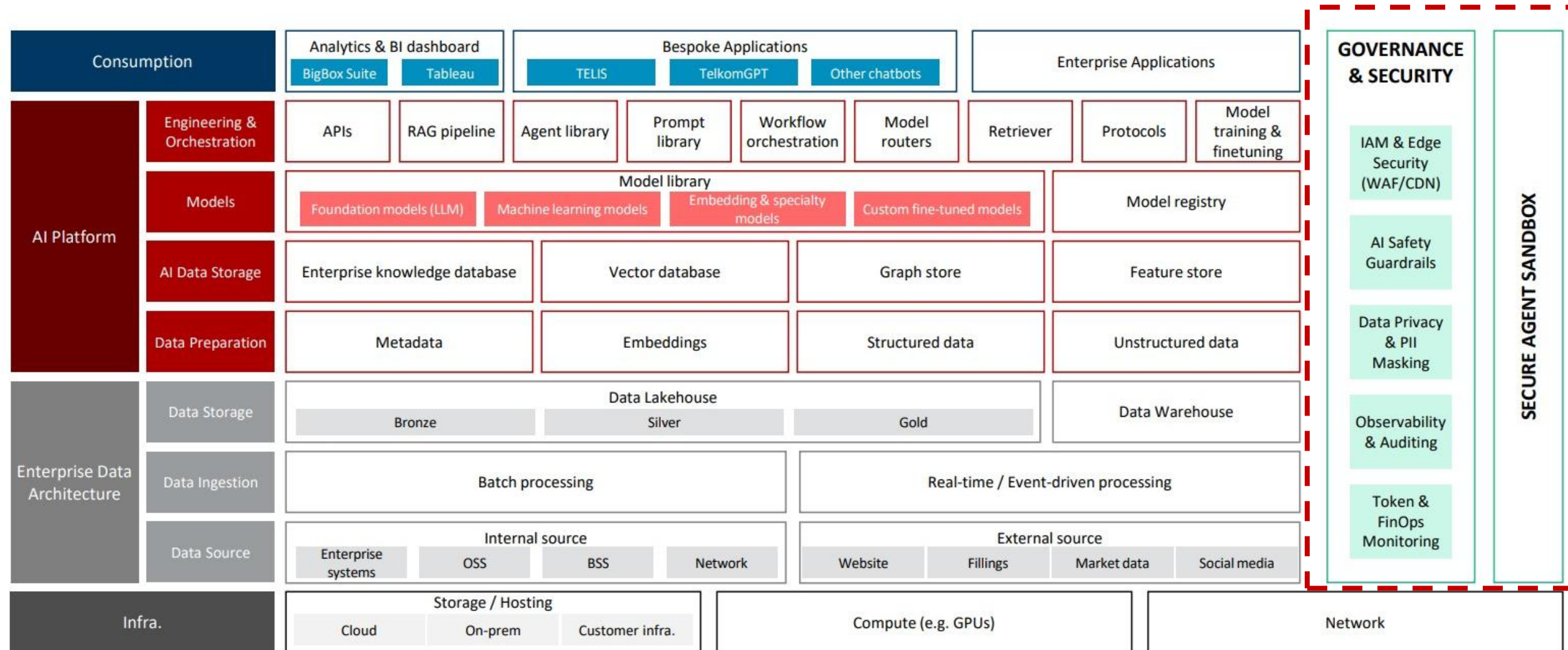
AI Sandbox System Flow

From Model Submission to Certification

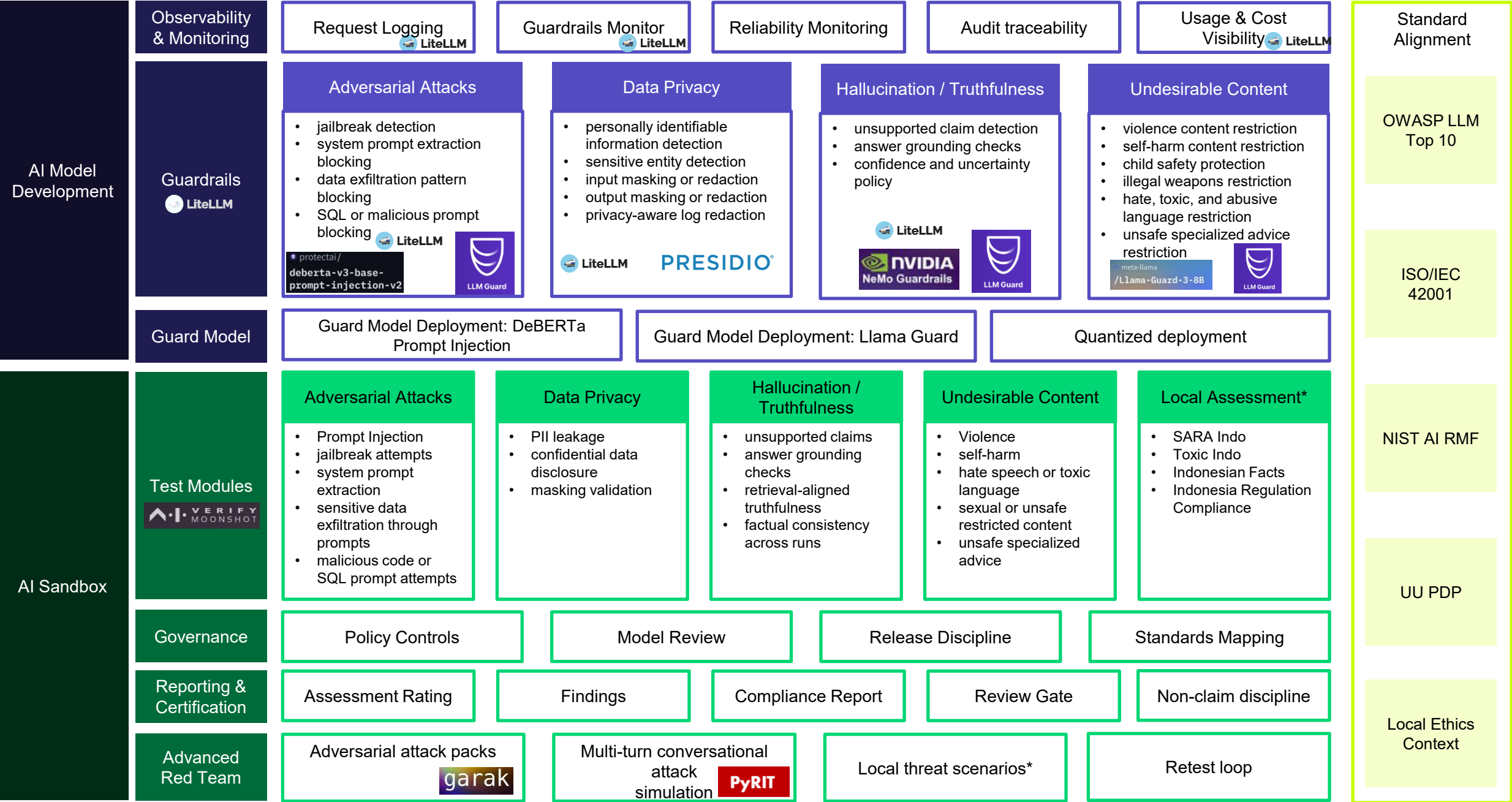


Based on enterprise AI testing, governance, and security practices.

The target AI platform architecture will enable Telkom to execute on all priority AI use cases



Security for AI Architecture



Roadmap SECURITY for AI

	Q1 2026 (Done) 	Q2 2026 	Q3 2026 	Q4 2026 	Future (2027) 
GOALS	Establish LiteLLM as the deployed Model gateway & Testing 1 Internal AI Model 	Operationalize the built-in LiteLLM Content Filter as the baseline protection layer	Operationalize the built-in LiteLLM Content Filter as the baseline protection layer	Implement structured expansion for privacy and prompt-attack handling	Strengthen truthfulness and grounding controls for higher-risk production paths
Observability& Monitoring	Deploy LiteLLM and track requests and tokens 	- Listed All models manage by DDP - Monitor All AI Usage in DDP 	- Add guardrail event logging - Link LiteLLM request and token telemetry	Make blocked or flagged events traceable per request	Add deeper data-exfiltration pattern blocking through LLM Guard
Guardrails	Testing LiteLLM Guardrails Feature 	Create LiteLLM Guardrails Setup	Establish baseline gateway using LiteLLM Content Filter, blocking baseline for: - prompt injection - denied medical, legal, and financial advice - harmful violence, self-harm, child safety, illegal weapons - toxic and abusive language - keyword blocking - pattern matching - block code execution - Prebuilt pattern PII masking	Add system prompt extraction blocking & data exfiltration pattern blocking (LLM Guard) Integrate Presidio PII detection	Operationalize NeMo Guardrails or Guardrails AI for runtime grounding Begin Indonesian localization tuning Improve handling of Bahasa slang, code-mixing, and local harm categories
Guard Model			Deploy DeBERTa Prompt Injection model	Deploy Llama Guard 3 as Guard Model	Quantized model by fine tuning security model
Model Testing	- Deploy AI Verify Moonshot as benchmark application baseline - Assess Telkom AI Using AI Verify Moonshot (Adversarial, Privacy, Hallucination) - Risk Register for Telkom AI Security Assessment 	Assess All Model sovereign for published catalogue: - Telkom AI (Qwen) - Telkom AI Instruct - Telkom AI Coder - Gemma 4	Re-assess model after implement guardrails	Re-assess model after implement guardrails Develop AI Sandbox as the first layer for internal model providers / fine tuning	Re-assess model after implement guardrails Develop AI Sandbox as the MVP for external model provider

LiteLLM Guardrails

LiteLLM Content Filter

[→ Show all \(22\)](#)

Built-in guardrails powered by LiteLLM. Zero latency, no external dependencies, no additional cost.

Denied Financial Advice

Detects requests for personalized financial advice, investment recommendations, or financial...

🟢 FI: 100% · 207 test cases

Insults & Personal Attacks

Detects insults, name-calling, and personal attacks directed at the chatbot, staff, or other people.

🟢 FI: 100% · 299 test cases

Denied Legal Advice

Detects requests for unauthorized legal advice, case analysis, or legal recommendations.

Denied Medical Advice

Detects requests for medical diagnosis, treatment recommendations, or health... advice.

Harmful Violence

Detects content related to violence, criminal planning, attacks, and violent threats.

Harmful Self-Harm

Detects content related to self-harm, suicide, and dangerous self-destructive behavior.

Harmful Child Safety

Detects content that could endanger child safety or exploit minors.

Harmful Illegal Weapons

Detects content related to illegal weapons manufacturing, distribution, or acquisition.

Bias: Gender

Detects gender-based discrimination, stereotypes, and biased language.

Bias: Racial

Detects racial discrimination, stereotypes, and racially biased content.

Listed default Guardrails in AI Gateway tools

Guardrail ID	↑↓	Name	↑↓	Provider	↑↓	Mode	↑↓	Default On	↑↓	Created At	▼	Updated At	↑↓	Actions
71e3c74...		Baseline Guard		litellm_content_filter		pre_call		Default Off		4/7/2026, 1:58:29 PM		4/7/2026, 2:28:49 PM		
8c2b7a7...		nsfw-racial-bias-filter-basic		litellm_content_filter		pre_call		Default Off		4/7/2026, 1:50:24 PM		4/7/2026, 1:50:24 PM		
e03e1f2...		nsfw-child-safety-filter-basic		litellm_content_filter		pre_call		Default Off		4/7/2026, 1:50:24 PM		4/7/2026, 1:50:24 PM		
3ca070b...		nsfw-self-harm-filter-basic		litellm_content_filter		pre_call		Default Off		4/7/2026, 1:50:24 PM		4/7/2026, 1:50:24 PM		
d385f75...		nsfw-content-filter-english-only		litellm_content_filter		pre_call		Default Off		4/7/2026, 1:50:24 PM		4/7/2026, 1:50:24 PM		
bdF4200...		Prompt Injection: Jailbreak		litellm_content_filter		pre_call		Default Off		4/7/2026, 1:48:42 PM		4/7/2026, 1:48:42 PM		
c345137...		Guardrails AI		guardrails_ai		pre_call		Default Off		4/7/2026, 1:46:16 PM		4/7/2026, 1:46:16 PM		
86795cc...		Pattern Matching		Presidio PII		pre_call		Default Off		4/7/2026, 1:45:23 PM		4/7/2026, 1:45:23 PM		
7b943df...		Presidio PII		Presidio PII		pre_call		Default Off		4/7/2026, 1:44:37 PM		4/7/2026, 1:44:37 PM		
d9690cc...		Harmful Violence		litellm_content_filter		pre_call		Default Off		4/6/2026, 10:04:06 AM		4/6/2026, 10:04:06 AM		
27d254e...		PAM GR Test		litellm_content_filter		pre_call		Default On		4/6/2026, 9:59:58 AM		4/6/2026, 10:03:03 AM		

Guardrails set up menu

**THANK YOU.
LET'S COLLABORATE!**

